

WEEK 1 PROJECT REPORT

Data Acquisition, Cleaning, and Exploratory Analysis

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1. Objective

Complete the data acquisition, cleaning, preprocessing, and exploratory analysis workflow using Python.

2. Dataset

The Wine Recognition dataset contains 178 observations, 13 numerical chemical features, and three target classes. It was loaded programmatically using Scikit-learn.

3. Data Cleaning

- Rows before cleaning: 178.
- Missing values: 0.
- Duplicate rows: 0.
- No rows required removal in this analysis.
- Data types and descriptive statistics were inspected.

4. Exploratory Analysis

Summary statistics and visualizations were used to understand class distribution, feature distributions, relationships, and correlations.

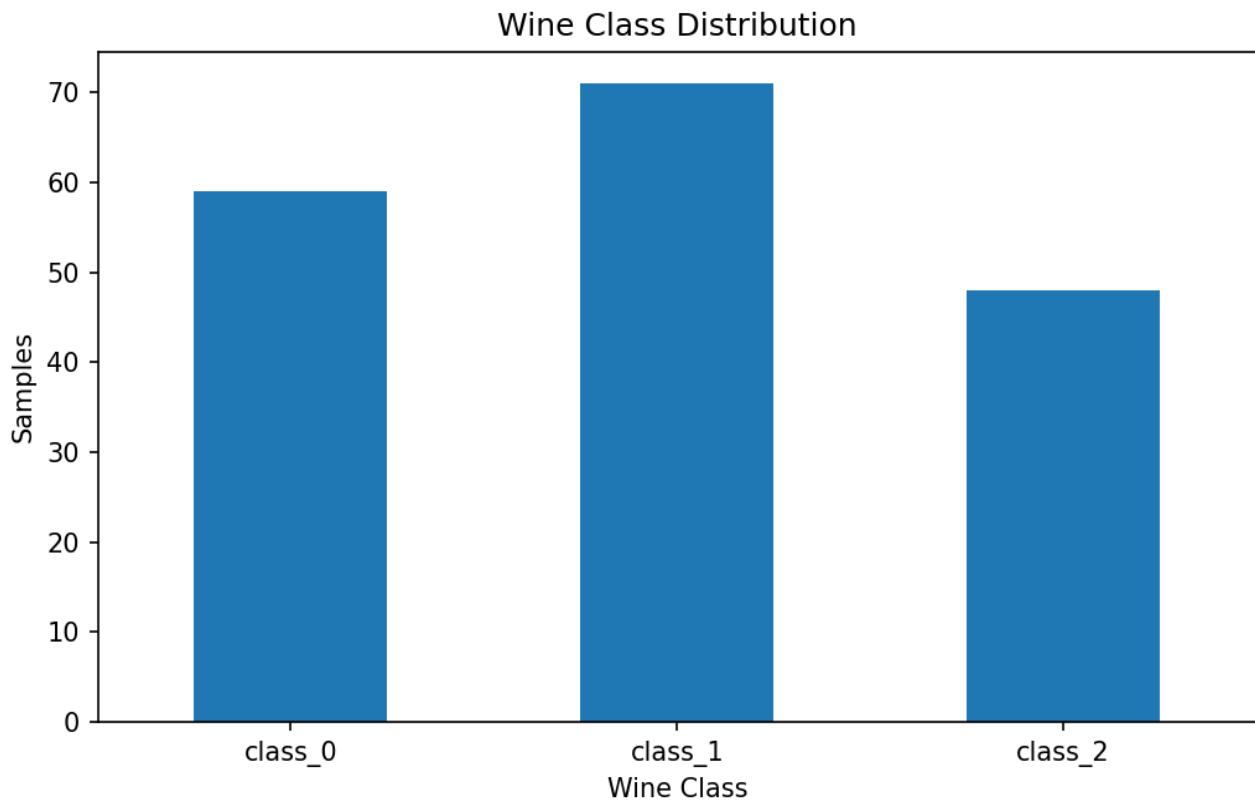
5. Key Insights

- The three wine classes have different sample counts.
- No missing values or duplicate rows were detected.
- Alcohol varies across observations and classes.
- Several chemical features show relationships useful for further analysis.

6. Conclusion

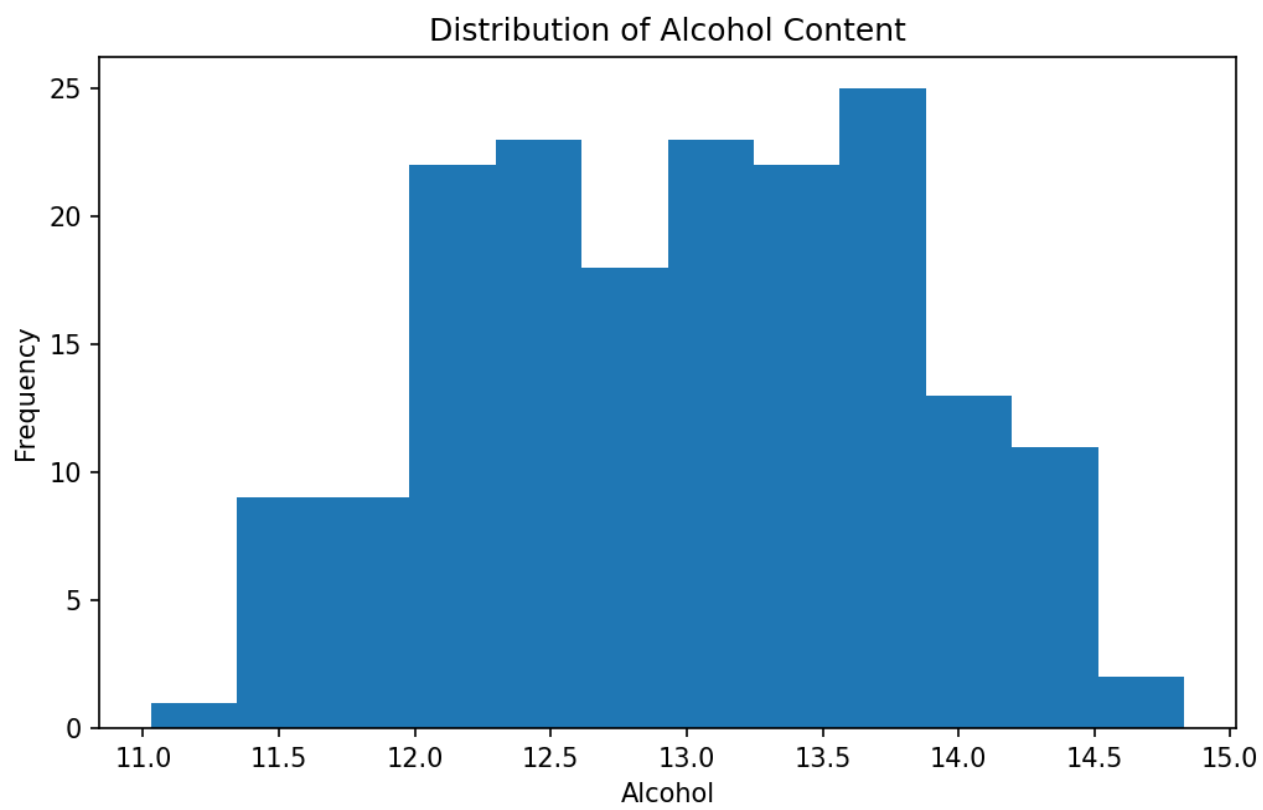
The dataset is suitable for downstream visualization, statistical testing, and classification modeling.

7. Class Distribution



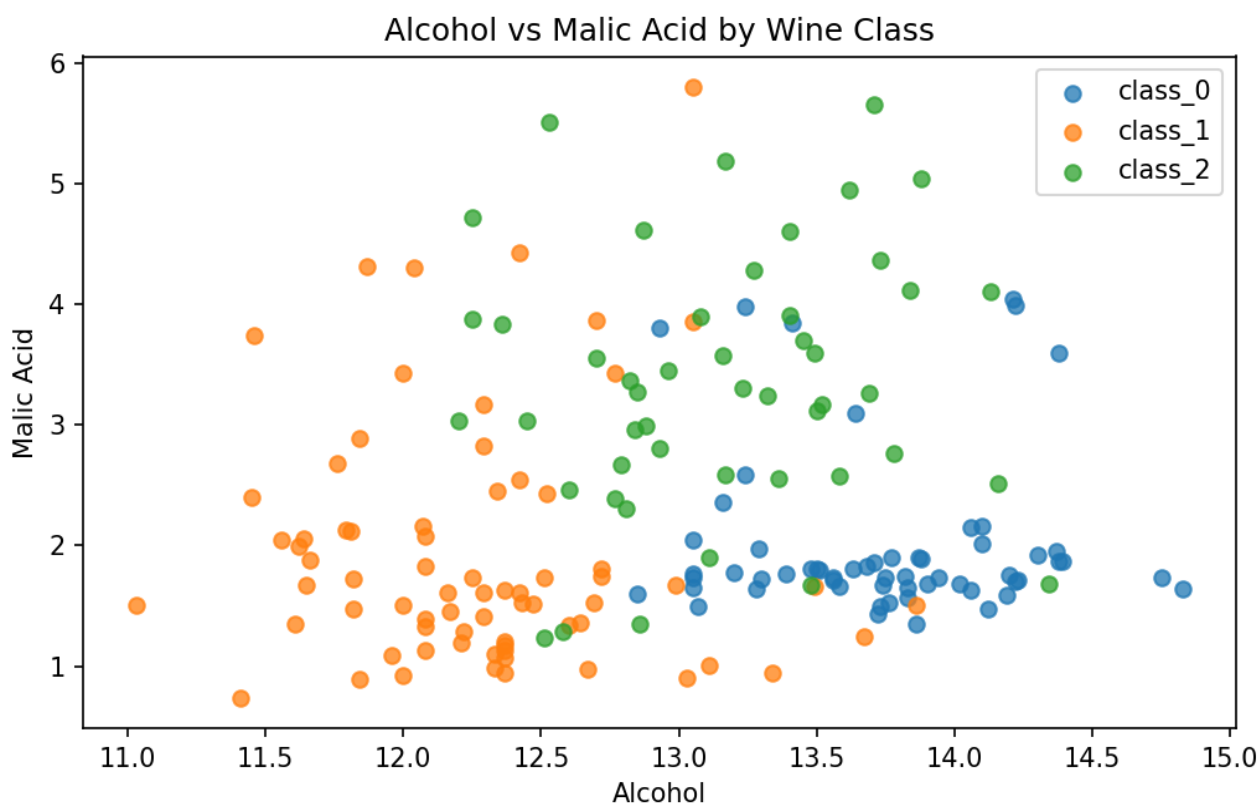
The chart shows the number of observations in each wine class.

8. Alcohol Distribution

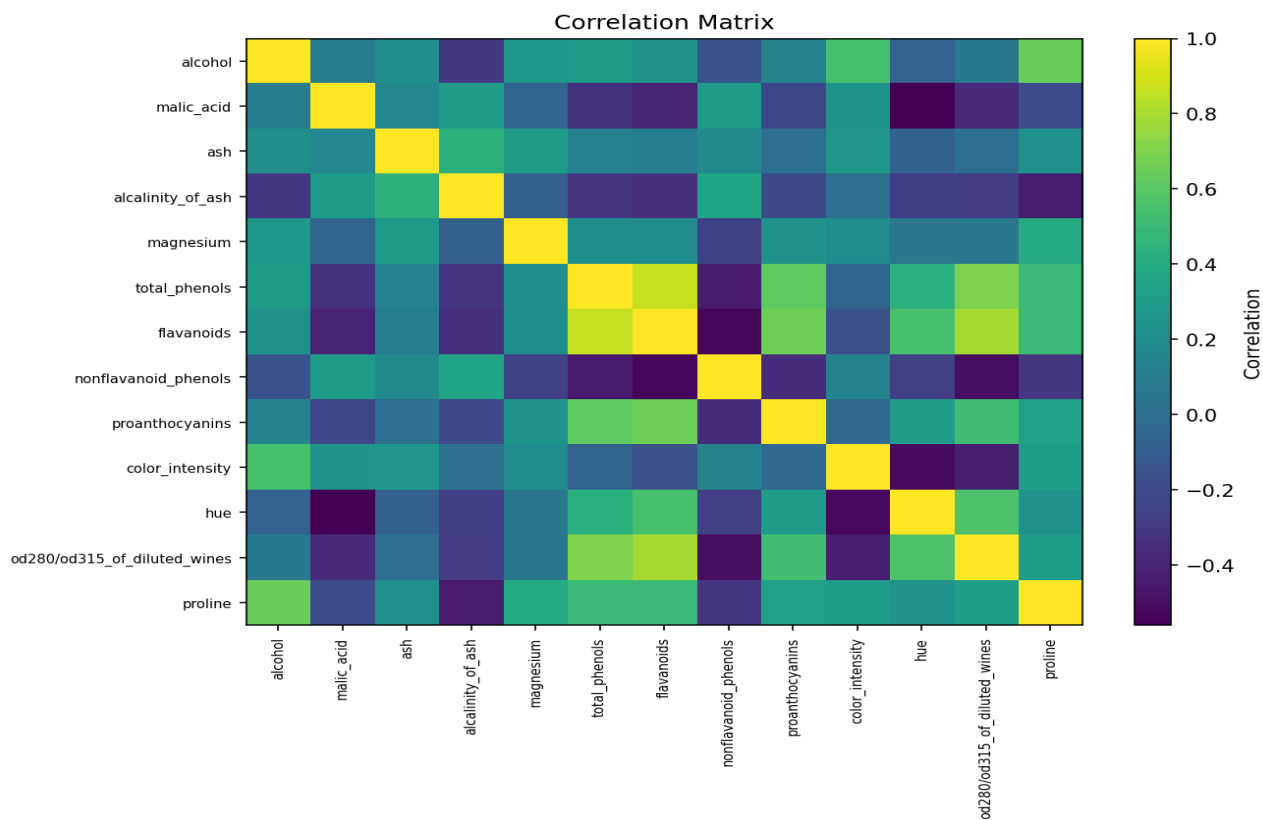


The histogram shows the distribution of alcohol content.

9. Alcohol vs Malic Acid



10. Correlation Matrix



The matrix summarizes pairwise linear correlations.